

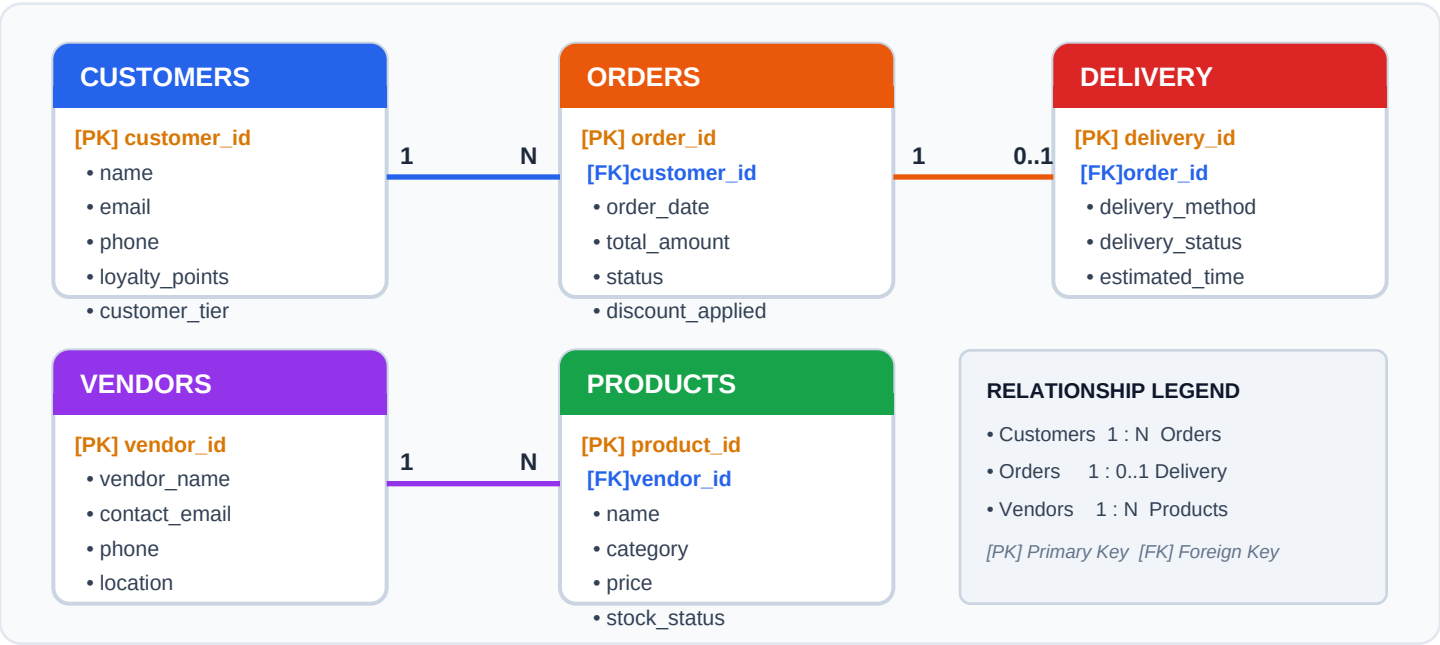
GrocerCart — Entity-Relationship (ER) Diagram

Oracle XE 11g Database Architecture & Data Model Documentation

1. System Overview & Relational Model

GrocerCart is an e-commerce grocery management system built on Oracle XE 11g. The relational model consists of 5 primary entity sets (**C**ustomers, **V**endors, **P**roducts, **O**rders, and **D**elivery) designed with full normalization, foreign key constraints, and PL/SQL triggers.

2. Visual Entity-Relationship Diagram



3. Detailed Entity Dictionary & Schema

Table: CUSTOMERS — Stores customer profile, contact, address, and loyalty rewards data.

Column Name	Data Type	Constraints	Description
customer_id	NUMBER	PK	Primary Key, auto-gen via seq_customers
name	VARCHAR2(100)	NOT NULL	Customer full name
email	VARCHAR2(150)	UNIQUE	Customer email address
phone	VARCHAR2(30)	—	Contact telephone number
address	VARCHAR2(500)	—	Physical delivery address
preferences	VARCHAR2(300)	—	Dietary & shopping preferences
loyalty_points	NUMBER	DEFAULT 0	Accumulated loyalty reward points
customer_tier	VARCHAR2(20)	CHECK	Bronze / Silver / Gold / Platinum

Table: VENDORS — *Stores suppliers and produce vendors offering inventory items.*

Column Name	Data Type	Constraints	Description
vendor_id	NUMBER	PK	Primary Key, auto-gen via seq_vendors
vendor_name	VARCHAR2(150)	NOT NULL	Vendor business name
contact_email	VARCHAR2(150)	—	Vendor contact email
phone	VARCHAR2(30)	—	Vendor phone number
location	VARCHAR2(200)	—	Operating location / address

Table: PRODUCTS — *Stores catalog products linked to their supplying vendors.*

Column Name	Data Type	Constraints	Description
product_id	NUMBER	PK	Primary Key, auto-gen via seq_products
vendor_id	NUMBER	FK	Foreign Key -> Vendors(vendor_id)
name	VARCHAR2(150)	NOT NULL	Product / Item name
category	VARCHAR2(100)	—	Category (Fruits, Dairy, Grains, etc.)
price	NUMBER(10,2)	NOT NULL, CHECK	Price (\$), must be >= 0
sustainability_tag	VARCHAR2(100)	—	Eco tag (Organic, Fair-Trade, etc.)
stock_status	VARCHAR2(20)	CHECK	In Stock / Low Stock / Out of Stock

Table: ORDERS — *Stores customer orders with order total, status, and discount.*

Column Name	Data Type	Constraints	Description
order_id	NUMBER	PK	Primary Key, auto-gen via seq_orders
customer_id	NUMBER	FK	Foreign Key -> Customers(customer_id)
order_date	DATE	DEFAULT SYSDATE	Date and time order placed
total_amount	NUMBER(10,2)	DEFAULT 0	Total purchase amount (\$)
status	VARCHAR2(20)	CHECK	pending / processing / shipped / delivered / cancelled
discount_applied	NUMBER(10,2)	DEFAULT 0, CHECK	Discount applied (\$), must be >= 0

Table: DELIVERY — *Stores delivery dispatch method, status, and estimated arrival date.*

Column Name	Data Type	Constraints	Description
delivery_id	NUMBER	PK	Primary Key, auto-gen via seq_delivery
order_id	NUMBER	FK, CASCADE	Foreign Key -> Orders(order_id) ON DELETE CASCADE
delivery_method	VARCHAR2(100)	—	Standard / Express / Pickup / Same Day
delivery_status	VARCHAR2(20)	CHECK	scheduled / in-transit / delivered / failed / ready
estimated_time	DATE	—	Estimated arrival date/time

4. Relational Constraints & Cardinalities

Parent Table	Child Table	Foreign Key	Cardinality	Delete Rule
Vendors	Products	vendor_id	1 : N (One-to-Many)	RESTRICT
Customers	Orders	customer_id	1 : N (One-to-Many)	RESTRICT
Orders	Delivery	order_id	1 : 0..1 (One-to-One)	CASCADE